

NLP202 - NER system

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1 Introduction

This report details the implementation and evaluation of a Named Entity Recognition (NER) system using a BIO tagger for the CoNLL 2003 shared task. The system identifies four entity types: people (PER), locations (LOC), organizations (ORG), and miscellaneous entities (MISC). I implemented a linear model with the Viterbi algorithm for decoding, trained using the structured perceptron with stochastic subgradient descent (SSGD) and early stopping. The assignment required evaluating the model with both a limited feature set (features 1-4) and the full feature set, as well as exploring structured SVM training with modified cost functions. The system employs Viterbi decoding and structured learning to enhance accuracy and is evaluated using precision, recall, and F1-score.

2 Dataset

The dataset follows the CoNLL-2003 format and consists of:

- **ner.train** - Training dataset
- **ner.dev** - Development dataset for validation
- **ner.test** - Test dataset for final evaluation
- **gazetteer.txt** - List of known location names for feature enhancement

Each line in the dataset includes:

1. Word
2. Part-of-Speech (POS) tag
3. Named Entity (BIO tagging format)

3 Feature Engineering

To improve entity recognition, the following features were implemented:

- **Current word** (W_i)
- **Lowercased word** (O_i)
- **POS tag** (P_i)
- **Previous tag** (T_{i-1})
- **Word shape** (S_i): Character transformations to capture capitalization patterns.
- **Contextual features**: Incorporating the previous and next word/POS tag.
- **Gazetteer match** (GAZ_i): Checking if the word exists in `gazetteer.txt`.
- **Capitalization flag** (CAP_i): Detects if the word starts with a capital letter.
- **Prefix-based features** (PRE_i): Prefixes of lengths 1-4 for the current word.

4 Model Implementation

The system consists of the following steps:

1. **Data Loading**: Parsing the dataset to extract words, POS tags, and named entity labels.
2. **Feature Extraction**: Generating feature vectors for each word in a sentence.
3. **Viterbi Algorithm**: Implementing a structured decoder for sequence prediction.
4. **Structured Perceptron Training**: Iterative weight updates based on prediction errors.
5. **Evaluation**: Computing Precision, Recall, and F1-score.

5 Training Details

The model was trained with the following specifications:

- **Algorithm**: Structured Perceptron
- **Epochs**: 20
- **Training Data**: `ner.train`
- **Validation Data**: `ner.dev`
- **Test Data**: `ner.test`
- **Optimizer**: Stochastic Gradient Descent (SGD)

5.1 Parts 1-3

This section corresponds to the code in the second document, which implements a Named Entity Recognition (NER) model using basic features:

- Word and tag combination ($W=\text{word}+T=\text{tag}$)
- Previous tag and current tag transition ($T-1=\text{prev_tag}+T=\text{tag}$)
- Lowercase word and tag ($O=\text{word.lower()}+T=\text{tag}$)
- POS tag and tag ($P=\text{pos}+T=\text{tag}$)

The model is trained using a structured perceptron with the Viterbi algorithm for decoding. The training runs for 20 epochs, and the performance is evaluated on the development (dev) and test sets.

5.2 Parts 4-8

- Part 4: Gazetteer Features - Checks if the word is in a gazetteer list ($GAZ=\text{True}+T=\text{tag}$).
- Part 5: Word Shape Features - Identifies patterns like all caps, capitalized, lowercase, or numbers (e.g., $SHAPE=\text{ALL_CAPS}+T=\text{tag}$).
- Part 6: Prefix and Suffix Features - Extracts prefixes and suffixes of lengths 2, 3, and 4 (e.g., $PREFIX=\text{word[:length]}+T=\text{tag}$).
- Part 7: Context Features - Includes previous and next words and POS tags (e.g., $W-1=\text{prev_word}+T=\text{tag}$).
- Part 8: Character N-gram Features - Extracts character n-grams of lengths 2 and 3 (e.g., $CHAR_length=\text{word[j:j+length]}+T=\text{tag}$).

6 Results

For the basic features, the best dev F1 score was **0.7448** at Epoch 13, and the weights from this epoch were used for evaluation.

Table 1: Dev Set Performance (Basic Features, Parts 1-3)

Label	Precision	Recall	F1-Score	Support
B-MISC	0.00	0.00	0.00	4
I-LOC	0.15	0.84	0.25	2094
I-MISC	0.22	0.52	0.31	1264
I-ORG	0.35	0.44	0.39	2092
I-PER	0.34	0.32	0.33	3149
O	0.99	0.72	0.83	42974
Summary Metrics				
Accuracy	0.68			
Macro Avg	0.34	0.47	0.35	51577
Weighted Avg	0.87	0.68	0.74	51577

Table 2: Test Set Performance (Basic Features, Parts 1-3)

Label	Precision	Recall	F1-Score	Support
B-LOC	0.00	0.00	0.00	6
B-MISC	0.00	0.00	0.00	9
B-ORG	0.00	0.00	0.00	5
I-LOC	0.14	0.82	0.25	1919
I-MISC	0.20	0.51	0.28	909
I-ORG	0.40	0.43	0.41	2491
I-PER	0.26	0.20	0.23	2773
O	0.99	0.73	0.84	38553
Summary Metrics				
Accuracy	0.68			
Macro Avg	0.25	0.34	0.25	46665
Weighted Avg	0.86	0.68	0.75	46665

Observations

- The model struggles with B- tags (e.g., B-MISC, B-LOC, B-ORG), with F1-scores of 0.00 due to low precision and recall.
- Performance on I- tags is better, especially for I-LOC (F1: 0.25) and I-ORG (F1: 0.41 on test), but still relatively low.
- The O tag (non-entity) has a high F1-score (0.83–0.84), indicating the model is biased towards predicting non-entities.
- The weighted average F1-score is 0.74 (dev) and 0.75 (test), but the macro average F1-score is much lower (0.35 dev, 0.25 test), showing poor performance on minority classes.

As for the advanced features, the best dev F1 score was **0.9748** at Epoch 18, and these weights were used for evaluation.

Table 3: Sample CoNLL Output (Basic Features, Parts 1-3)

Word	POS	Gold Tag	Predicted Tag
SOCCER	NN	O	O
-	:	O	O
JAPAN	NNP	I-LOC	I-LOC
GET	VB	O	I-ORG
LUCKY	NNP	O	I-LOC
WIN	NNP	O	I-LOC
,	,	O	O
CHINA	NNP	I-PER	I-LOC
IN	IN	O	O
SURPRISE	DT	O	O
DEFEAT	NN	O	I-LOC
.	.	O	O

Table 4: Dev Set Performance (Advanced Features, Parts 4-8)

Label	Precision	Recall	F1-Score	Support
B-LOC	0.00	0.00	0.00	0
B-MISC	0.20	0.50	0.29	4
I-LOC	0.86	0.90	0.88	2094
I-MISC	0.93	0.78	0.85	1264
I-ORG	0.87	0.82	0.84	2092
I-PER	0.92	0.92	0.92	3149
O	0.99	1.00	0.99	42974
Summary Metrics				
Accuracy	0.98			
Macro Avg	0.68	0.70	0.68	51577
Weighted Avg	0.97	0.98	0.97	51577

Observations

- The advanced features significantly improve performance. The weighted average F1-score jumps to 0.97 (dev) and 0.95 (test), compared to 0.74 and 0.75 with basic features.
- The I- tags see substantial improvements: I-PER F1-score increases from 0.33 to 0.92 (dev), and I-LOC from 0.25 to 0.88 (dev).
- The B- tags still struggle, with F1-scores of 0.00 for B-LOC and B-ORG, though B-MISC improves slightly (0.29 on dev, 0.12 on test).
- The O tag maintains a high F1-score (0.99), and the macro average F1-score rises to 0.68 (dev) and 0.54 (test), indicating better handling of minority classes.
- The gazetteer, word shape, prefix/suffix, context, and character n-gram features likely contribute to better disambiguation of entities, especially for I- tags.

Table 5: Test Set Performance (Advanced Features, Parts 4-8)

Label	Precision	Recall	F1-Score	Support
B-LOC	0.00	0.00	0.00	6
B-MISC	0.14	0.11	0.12	9
B-ORG	0.00	0.00	0.00	5
I-LOC	0.74	0.85	0.79	1919
I-MISC	0.81	0.74	0.77	909
I-ORG	0.77	0.74	0.75	2491
I-PER	0.86	0.86	0.86	2773
O	0.99	0.99	0.99	38553
Summary Metrics				
Accuracy	0.95			
Macro Avg	0.54	0.54	0.54	46665
Weighted Avg	0.96	0.95	0.95	46665

Impact of Advanced Features

- **Gazetteer (Part 4):** Helps identify known entities (e.g., locations or organizations in the gazetteer), improving recall for I-LOC and I-ORG.
- **Word Shape (Part 5):** Captures patterns like capitalization, which is crucial for proper nouns (e.g., I-PER improves to 0.92 F1 on dev).
- **Prefix/Suffix (Part 6):** Provides morphological clues, aiding in entity boundary detection.
- **Context (Part 7):** Previous and next words/POS tags improve tag transitions, reducing errors like tagging “GET” as I-ORG.
- **Character N-grams (Part 8):** Adds robustness to unseen words by capturing subword patterns, benefiting I-MISC and I-ORG.

Comparison of Basic vs. Advanced Features

The advanced features lead to a significant performance boost:

- **Dev F1 (Weighted Avg):** 0.74 (Basic) → 0.97 (Advanced)
- **Test F1 (Weighted Avg):** 0.75 (Basic) → 0.95 (Advanced)
- **Macro F1 (Dev):** 0.35 (Basic) → 0.68 (Advanced)
- **Macro F1 (Test):** 0.25 (Basic) → 0.54 (Advanced)

The improvements are most pronounced for I- tags, though B- tags remain a challenge, likely due to their rarity in the dataset (low support values).

Model Variant	Precision	Recall	F1-score
Baseline (Basic Features)	86.0%	68.0%	75.0%
Full Feature Model (Advanced Features)	96.0%	95.0%	95.0%

Table 6: NER Model Performance Comparison (Test Set, Weighted Average Metrics)

Conclusion

The basic features provide a reasonable baseline for NER but struggle with minority classes and complex entity patterns. The advanced features dramatically improve performance by incorporating richer contextual and lexical information, making the model more robust and accurate. However, further improvements are needed for B- tags, possibly through better handling of rare entities or additional training data.